

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Dec 2023

Course Code: CHY101

Course: ENGINEERING CHEMISTRY

QP No. :UD158-1

Duration: 3 hours

Max. Marks:100

PART- A

Answer all the questions

10 x 2 = 20 Marks

1. The use of hard water for a long time resulted the scale formation. Washing with 5-10% HCl did not remove the scales completely. Explain the reason and suggest a suitable method to remove the scale.
2. Water supplied to a boiler industry was acidic in nature. Boiler industry want to perform internal treatment to avoid scale formation. Suggest a suitable internal treatment method and provide the reason.
3. A village people usually collect water from a nearby pond for domestic usage. Whenever there was a heavy rain in the surrounding area, the water becomes turbid. The people found difficulty in removing the turbidity by keeping the water undisturbed. Why the water remains turbid even after standing for few days? Suggest a method to make the water clean from the turbidity.
4. A copper and silver wire were placed in silver nitrate and copper sulphate aqueous solution, respectively for a day. What would be the observation? Justify.
5. Fuel cells (a) are free from vibrations, heat transfer and thermal pollution and (b) can produce drinking water of potable quality. (i) both (a) and (b) are true, (ii) (a) true (b) false, (iii) (a) false, (b) true and (iv) both (a) and (b) are false.
6. Iron embedded cement concrete blocks were made for construction purpose. After few months, the workers noticed that the cement concrete is peeled off from the iron rods in some of the blocks without any morphological change. In some other blocks, cements volume increased and appeared as

bulged with porous structure. Explain the reason for the above observation on the basis of corrosion.

7. Dimethyl ether (CH_3OCH_3) and ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) has same molecular formulae ($\text{C}_2\text{H}_6\text{O}$) but showed completely different boiling point, why?
8. For two different lubricants (A and B), the saponification values were observed to be 30 and 0.5 respectively. With logical reasoning, identify the mineral oil and vegetable oil among them?
9. Comment on the strength of composite prepared using dispersion particles with (a) 100 nm and (b) 100 μm spherical particles.
10. Fine dust particles generated upon the movement of vehicles easily attached on any surfaces but same materials at bigger size did not attach. Why?

PART- B

Answer all the questions

4 x 15 = 60 Marks

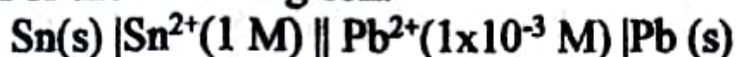
11. (a) Two water sample (i and ii) from different locations are collected and tested. The water analysis report of the two sample per 1L of water is given below.
 - (i) MgCl_2 (120 mg), $\text{Ca}(\text{HCO}_3)_2$ (150 mg), NaNO_3 (100 mg), FeSO_4 (150 mg) and K_2SO_4 (100 mg).
 - (ii) NaCl (50 mg), MgSO_4 (100 mg), KCl (75 mg) and CaSO_4 (50 mg).Calculate the temporary, permanent and total hardness of both the sample. Can you soften both the water sample using zeolite? Explain. (Atomic weight of $\text{Mg} = 24.3$, $\text{Ca} = 40$, $\text{Na} = 23$, $\text{Fe} = 55.8$, $\text{K} = 39$, $\text{H} = 1$, $\text{Cl} = 35.45$, $\text{C} = 12$, $\text{O} = 16$, $\text{N} = 14$ and $\text{S} = 32$). (8)
- (b) The availability of good water for domestic usage is limited across the globe and worse especially in the middle east. Suggest suitable methods to purify the available high salt containing water and discuss the mechanism of water purification with the diagram. (7)

(OR)

12. (a) 250 mL of water is boiled for half an hour and filtered. The filtrate is made up to 250 mL with distilled water. 50 mL of this water required 4 mL of 0.02 M EDTA solution with an NH_4OH - NH_4Cl buffer and EBT indicator. Calculate permanent hardness of the sample. Also calculate the number of liters of NaCl (having a strength of 100 g/L) required to regenerate a zeolite softener if 1000 L of this water softened. (7)

- (b) A water-based company is looking for developing a technology to purify the water to obtain the water with very low residual hardness. Which method would be more suitable to remove the dissolved ions completely from the water? Explain neatly with the help of diagram and chemical equation. (8)

13. (a) For the following cell.



Calculate the emf, comment on the spontaneity of the cell and also calculate the free energy change for the reaction.

The standard reduction potential values of Sn^{2+}/Sn and Pb^{2+}/Pb are -0.14 V and -0.13 V respectively.

Also, what will happen if the concentration of Pb^{2+} solution is changed to 0.2 M ? (8)

- (b) Discuss the mechanism of lead-acid and lithium-ion batteries charging and discharging along with its advantages. (7)

(OR)

14. (a) An iron rod was exposed to acidic, alkaline, freezing ice (-40°C) and nitrogen environment. Is the corrosion of iron different in different environment? Justify with necessary reactions/schemes. (8)

- (b) An oil industry used ship and iron pipe for transporting oil in the sea and land, respectively. But the disintegration/breaking of iron structure/pipe due to corrosion was a major issue. Suggest appropriate preventive technology to protect the iron from corrosion and explain the mechanism. (7)

15. (a) Justify the following statements:

(i) Although CO and N_2 has same atomic weight (28), CO has higher boiling point compared to N_2 .

(ii) Water exhibits a melting point of 0°C and boiling point of 100°C , while hydrogen sulphide exhibits melting point and boiling point of -82°C and -60°C respectively.

(iii) Polyethylene-based carry bag is losing the strength above 100°C while a Teflon block is stable even above 300°C . (8)

- (b) Suggest suitable lubricant for the following condition: A heavy load bearing condition such as railway tracks and a low-speed low load condition such as a cycle chain. Briefly discuss about the suggested lubricant choosing appropriate examples. (7)

(OR)

16. (a) Two oil samples with different thickness were given to you for measuring the viscosity. Suggest a suitable technique for measuring the viscosity and elaborately discuss. Do viscosity affects its applications. (8)

- (b) Define and explain the significance of cloud point, pour point and steam emulsion number. (7)

17. Three batches of fiber reinforced composite materials were prepared that showed the following properties, Batch A: the physical properties of the composite were isotropic, but with minimal reinforcement, Batch B: anisotropic behavior but the reinforcement was not still, as expected and Batch C: excellent mechanical strength in two dimensions (anisotropic). Explain the observations based on the nature of composites.

(OR)

18. (a) Demonstrate wet-chemical and physical vapor deposition method of nanoparticles synthesis. (6)

- (b) Nanoparticles are potential materials for various applications. Discuss the applications of silver nanoparticles for water purification, gold nanoparticles for anti-cancer activity and TiO_2 for self-cleaning windows. (9)

PART -C

Answer the following

1 x 20 = 20 Marks

19. (a) A hotel management used bleaching powder as disinfectant for cleaning the swimming pool. After the treatment, people started complaining skin and eye irritation while taking bath in the swimming pool. What would be reason for the skin and eye irritation? Discuss the mechanism disinfection action and method to prevent skin/eye irritation by taking bath in the swimming pool. (6)

- (b) A small-scale industrial person wants to make the insulative polymer toys more attractive by depositing metal ions such as copper, silver and gold. Suggest a suitable method for metal deposition and explain the mechanism. (5)

- (c) Invoking the intermolecular interactions, explain the following observations; (i) floating of a butter paper over water surface and (ii) aggregation of nanoparticles and prevention of nanoparticles aggregation using stabilizing agents. (9)

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